

- IMSC 2025 -

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M. Knežević, SRB

PROBLEMS

1. We say that a positive real number a is *verrrh* if there exists an infinite sequence $a_1, a_2, a_3, \dots \in (0, a)$ such that, for every n , the points $a_1, a_2, \dots, a_n$ divide the segment $[0, a]$ into subsegments whose lengths are at most $\frac{1}{n}$. Find

$$\sup\{a \mid a \text{ is verrrh}\}.$$

2. For real numbers $-1 < x_1, x_2, \dots, x_n < 1$, such that $x_1 + x_2 + \dots + x_n = 0$, prove that the following holds:

$$\sum_{i=1}^n \sum_{j=1}^n x_i x_j \sqrt{1 - x_i^2 x_j^2} \leq 0.$$

Determine the conditions under which equality is achieved.

3. [IND PTST 23] Let $\mathbb{R}_+$ be the set of all positive real numbers. Find all functions $f: \mathbb{R}_+ \rightarrow \mathbb{R}_+$ satisfying

$$f(x + y^2 f(x^2)) = f(xy)^2 + f(x),$$

for all $x, y \in \mathbb{R}_+$.

4. [IR TST 24] Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that, for any real numbers x and y , the equality holds:

$$f(yf(x) + f(x)f(y)) = xf(y) + f(xy).$$

5. Let $\mathbb{R}$ denote the set of all real numbers. Find all pairs of functions

$$f: \mathbb{R} \rightarrow \mathbb{R} \quad \text{and} \quad h: \mathbb{R}^2 \rightarrow \mathbb{R}$$

such that, for all $x, y, z \in \mathbb{R}$, the following holds:

$$f(x + y - z)^2 = f(xy) + h(x + y + z, xy + yz + zx).$$

6. [IR PTST 23] Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that, for all $x, y \in \mathbb{N}$, the following holds:

$$0 \leq y + f(x) - f^{f(y)}(x) \leq 1,$$

where

$$f^n(x) = \underbrace{f(f(\dots f(x) \dots))}_{n \text{ times}}, \quad n \in \mathbb{N}.$$

7. [IND IMOTC PTST 24] Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$xf(xf(y) + yf(x)) = x^2 f(y) + yf(x)^2,$$

for all real numbers x and y .

8. Prove that for every natural number n there exist complex numbers $\omega_1, \omega_2, \dots, \omega_n$, lying on the unit circle, such that

$$\left| \sum_{j=1}^n \omega_j \right| = \left| \sum_{j=1}^n \omega_j^2 \right| = n - 1.$$

9. Find all functions $f : \mathbb{C} \rightarrow \mathbb{C}$ such that, for all complex numbers x, y , the following holds:

$$f(f(x) + yf(y)) = x + |y|^2.$$

10. Let $\mathbb{R}_+ = \{x \in \mathbb{R} \mid x > 0\}$ denote the set of positive real numbers. Find all pairs of functions $f, g : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that the following equalities hold:

$$f(g(x)) = f(x)g(x), \quad f(x) = x(1 + g(x))$$

and the sequence $g(x), g(g(x)), g(g(g(x))), \dots$ has finitely many distinct values, for every $x \in \mathbb{R}_+$.

11. Find all sequences $a_1, a_2, \dots$ of positive integers such that the expression

$$na_n - ma_m + 2a_m - 1$$

is divisible by $a_n + a_m - 1$, for all $n, m \in \mathbb{Z}$, $n, m \geq 1$.

12. Find all strictly increasing functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that the following holds:

$$f(f(n)) + f(f(m)) + f(m) = f(n)f(m), \text{ for all } m, n \in \mathbb{N},$$

where $\mathbb{N}$ is the set of natural numbers.